

Alt Carbon

AI / Engineering Intern — Take-Home Assignment

3 parts · All data is synthetic ·

Confidential — do not share

What this is

This assignment mirrors what our engineering team actually does: process messy field data, implement carbon removal calculations, build validation logic, and think about what could go wrong at scale. Most of the work is writing code.

You can use AI tools. We do too.

Background

Alt Carbon removes CO₂ through Enhanced Rock Weathering (ERW). We spread crushed silicate rock on agricultural fields. As rock weathers in soil, it reacts with dissolved CO₂ and converts it to dissolved bicarbonates — a stable carbon sink. We prove this happened by collecting soil samples and running lab analysis.

The data pipeline

1. **Field collection.** Teams collect soil from GPS-tagged points. Baseline samples before rock application, monitoring samples 6–12 months after. Each sample gets a barcode and is shipped to a lab.
2. **Lab analysis.** ICP-OES measures 150+ element concentrations. Key immobile tracers: Ti (titanium), Zr (zirconium) — these shouldn't change with weathering. Key mobile elements: Ca, Mg, Si — these increase if weathering happened.
3. **Sample pairing.** Each monitoring sample is matched to a nearby baseline by GPS proximity. The immobile tracer values should be similar between paired samples.
4. **CDR calculation.** Carbon Dioxide Removal is computed by comparing mobile element ratios between paired baseline and monitoring samples, using an immobile tracer as the reference frame.
5. **Verification.** A third-party registry audits the data and calculations before issuing carbon credits.

The CDR formula

This is the simplified calculation you'll implement. Ti is the immobile tracer.

Step 1 — Enrichment ratio per mobile element:

$$\text{Enrichment_Ca} = (\text{Ca_mon} / \text{Ti_mon}) - (\text{Ca_bl} / \text{Ti_bl})$$

Step 2 — Normalize to soil mass:

$$\Delta\text{Ca (ppm)} = \text{Enrichment_Ca} \times \text{Ti_mon}$$

Step 3 — Convert to moles CO₂ (each mole Ca or Mg from silicate weathering captures 2 moles CO₂):

$$\text{CDR_Ca} = (\Delta\text{Ca} / 40.08) \times 2 \quad (\text{molar mass Ca} = 40.08)$$

$$\text{CDR_Mg} = (\Delta\text{Mg} / 24.31) \times 2 \quad (\text{molar mass Mg} = 24.31)$$

Step 4 — Sum CDR_Ca + CDR_Mg, convert mol to grams ($\times 44.01$ g/mol CO₂), then to tonnes/ha:

$$\text{CDR (t CO}_2\text{/ha)} = (\text{CDR_Ca} + \text{CDR_Mg}) \times 44.01 \times \text{soil_mass} / 1e6$$

where soil_mass = 2600 tonnes/ha (0–30 cm depth)

This is simplified. Real calculations include bootstrap resampling, control correction, and uncertainty quantification. This version is sufficient for this assignment.

Assignment structure

Part	What	Time	Weight
1	Data Quality – find and flag issues	45 min	20%
2	CDR Pipeline – implement the full calculation	90 min	50%
3	Thinking – edge cases, improvements, scale	30 min	30%

Your Data

All synthetic. Two CSV files representing a fictional ERW project called Project Ganga — rice paddies in eastern India. Rock applied to treatment plots in October 2024.

samples.csv

sample_id	type	lat	lon	date	collector	barcode	plot_type
GNG-BL-001	baseline	23.4512	87.3201	2024-10-12	Sunil D	LB-24-4401	treatment
GNG-BL-002	baseline	23.4515	87.3198	2024-10-12	Sunil D	LB-24-4402	treatment
GNG-BL-003	baseline	23.4610	87.3305	2024-10-13	Meera J	LB-24-4403	control
GNG-BL-004	baseline	23.4614	87.3310	2024-10-13	Meera J	LB-24-4404	control
GNG-BL-005	baseline	0.0000	0.0000	2024-10-14	Sunil D	LB-24-4405	treatment
GNG-MON-001	monitoring	23.4518	87.3195	2025-05-22	Sunil D	LB-25-5501	treatment
GNG-MON-002	monitoring	23.4513	87.3200	2025-05-22	Sunil D	LB-25-5502	treatment
GNG-MON-003	monitoring	23.4890	87.3610	2025-05-23	Rahul P	LB-25-5503	treatment
GNG-MON-004	monitoring	23.4512	87.3201	2025-05-24	Sunil D		treatment
GNG-MON-005	monitoring	23.4612	87.3308	2025-05-25	Meera J	LB-25-5505	control
GNG-BL-006	baseline	23.4512	87.3201	2025-05-24		LB-25-5506	treatment
GNG-MON-006	monitoring	23.4617	87.3312	2025-05-25	Meera J	LB-25-5507	control

lab_results.csv

barcode	Ti_ppm	Zr_ppm	Ca_ppm	Mg_ppm	Si_ppm	status
LB-24-4401	3102	158	16780	5890	8430	complete
LB-24-4402	3045	154	16520	5710	8290	complete
LB-24-4403	3200	162	17100	6050	8610	complete
LB-24-4404	3150	160	16900	5980	8500	complete
LB-24-4405	3080	156	16650	5820	8350	complete
LB-25-5501	3110	157	22340	8760	11420	complete
LB-25-5502	3058	155	21890	8510	11100	complete
LB-25-5503	5240	412	48200	19100	26800	flagged
LB-25-5505	3190	161	17800	6350	9050	complete
LB-25-5506	3095	157	16700	5850	8400	complete
LB-25-5507	3170	160	19800	7350	10100	complete

LB-25-5508	3130	159	21500	8400	10950	complete
------------	------	-----	-------	------	-------	----------

Note: LB-25-5508 has no matching entry in samples.csv.

Part 1: Data Quality Checks

20% of total score · Deliverable: code + output + any comments/suggestions you might have

Write a script/program that loads both CSVs, joins them on barcode, and runs automated quality checks. Your script should print a report listing every issue it finds.

Required checks

Implement at least these validation rules:

1. **Missing fields.** Flag any sample with missing barcode, missing collector, or GPS coordinates of (0, 0).
2. **Orphan lab results.** Flag barcodes in lab_results that don't match any sample in the collection log (and vice versa).
3. **Baseline timing.** Flag any sample labeled "baseline" that was collected *after* the rock application date (assume rock was applied 2024-10-15 for treatment plots).
4. **Spatial outliers.** For treatment monitoring samples, compute the distance to the nearest treatment baseline. Flag any monitoring sample more than 500m from its nearest baseline. (Use the Haversine formula or a simple Euclidean approximation for lat/lon.)
5. **Tracer stability.** After joining with lab results, flag any sample where Ti deviates more than 20% from the mean Ti of its plot type (treatment or control).
6. **Lab status.** Flag any sample where lab status is "flagged."

Output format

Your script should print something like:

```
ISSUE  GNG-BL-005    missing_gps        GPS is (0,0)
ISSUE  GNG-MON-004   missing_barcode    No barcode recorded
ISSUE  GNG-BL-006    late_baseline      Baseline collected 2025-05-24, after
rock application
ISSUE  LB-25-5508    orphan_lab_result   No matching sample in collection log
...
```

Exact formatting is up to you. What matters is that the output is clear and every issue is categorized.

Not everything is broken. Some values may look unusual but are within normal scientific ranges. Flagging non-issues shows you didn't think about what the checks actually mean.

What we're looking for: Clean, readable code. Sensible thresholds. Checks that go beyond surface-level (e.g., catching the timing issue on BL-006 requires understanding what "baseline" means). Bonus: additional checks we didn't list.

Part 2: CDR Calculation Pipeline

50% of total score · Deliverable: code + output + any comments/suggestions you might have

Write a script/program that takes the two CSVs as input and outputs a CDR estimate for the project. This is the core of the assignment.

What your pipeline should do

Step 1: Pair baselines to monitoring samples

For each treatment monitoring sample, find the nearest treatment baseline sample by GPS distance. A valid pair requires the baseline to be within 500m of the monitoring sample. Each baseline can only be used once (1:1 matching — if two monitoring samples compete for the same baseline, the closer one wins).

Step 2: Validate pairs

Before computing CDR, each pair must pass these gates:

- **Tracer stability:** reject if Ti differs more than 20% between baseline and monitoring
- **Baseline timing:** reject if baseline was collected after rock application date
- **Lab status:** reject if either sample has status “flagged”
- **Plot type:** only compute CDR for treatment pairs (not control)

Step 3: Compute CDR for each valid pair

Use the formula from the Background section. Calculate CDR for Ca and Mg separately, then sum. Convert to tonnes CO₂/ha using soil_mass = 2600 t/ha.

Step 4: Compute project-level statistics

- Mean CDR across all valid pairs
- Standard deviation
- Number of valid pairs vs. total pairs attempted
- 95% confidence interval (use t-distribution if you know it, or note that you can’t compute one with < 3 samples)

Step 5: Compute control plot CDR

Also pair and compute CDR for control plots using the same method. Control plots should show ~0 CDR (no rock was applied). Report the control CDR separately — this is a sanity check. If control CDR is significantly non-zero, flag it.

Expected output

Your script should print a structured report. Something like:

```
=== PAIRING ===
Pair 1: GNG-BL-001 <-> GNG-MON-002  dist=72m  VALID
Pair 2: GNG-BL-002 <-> GNG-MON-001  dist=85m  VALID
Pair 3: GNG-BL-005 <-> GNG-MON-003  dist=???  REJECTED (invalid GPS on
baseline)
...

=== CDR RESULTS (Treatment) ===
Pair 1:  CDR_Ca = X.XX  CDR_Mg = X.XX  Total = X.XX t CO2/ha
...

=== PROJECT SUMMARY ===
Valid pairs: N / M
Mean CDR: X.XX t CO2/ha
Std dev: X.XX
95% CI: [X.XX, X.XX] or 'insufficient samples'

=== CONTROL CHECK ===
Control CDR: X.XX t CO2/ha  (expect ~0)
```

Traps in the data: Not all pairs are straightforward. BL-005 has zeroed GPS. BL-006 is labeled “baseline” but was collected 7 months after rock application. MON-003 is far from any baseline. LB-25-5503 has wildly anomalous Ti/Zr. The control plot shows non-zero CDR. A good pipeline catches all of these. A great pipeline explains why each was rejected.

Part 3: Thinking

Deliverable: written (in README or separate doc)

We want to see how you think, not how much you write. Here are rough questions to guide you in case you want to start somewhere, kindly note that these questions are not hard bound but just a guide for you to start thinking about your submission document.

You can also submit your understanding of the methodology and the CDR calculation steps that you did above.

1. **What would break at 10x scale?** This dataset has 12 samples. Imagine 5,000 samples across 8 projects. What parts of your pipeline would fail or become too slow? What would you redesign?
2. **Pairing strategy.** Your pipeline matches monitoring to nearest baseline by GPS. What are the failure modes of this approach? Can you think of a better pairing strategy that also considers geochemistry (e.g., tracer similarity)?
3. **The control plot problem.** If your control plot CDR came out to $\sim 0.3 \text{ t CO}_2/\text{ha}$ (non-trivially positive), what would that mean scientifically? What should the engineering team do about it? How does this affect the treatment CDR numbers?
4. **Validation trade-offs.** You implemented a 20% Ti deviation threshold. What happens if you make it stricter (10%)? Looser (30%)? How would you decide the right threshold in practice? Who should have the authority to change it?
5. **One more thing.** If you had another week to work on this, what would you add to the pipeline? Be specific — name one feature and sketch how you'd implement it.

What we're evaluating: Not the “right” answer — there isn't one. We want to see that you can reason about trade-offs, anticipate failure modes, and think beyond the immediate code. Strong answers are specific and reference your actual implementation.

How we'll score this

Criteria	Weight	What “great” looks like
Data Quality (Part 1)	20%	Catches all planted issues, no false flags, clean code, goes beyond required checks
CDR Pipeline (Part 2)	50%	Correct CDR math, handles all edge cases, rejects bad pairs with reasoning, good code structure

Thinking (Part 3)	30%	Your process, ideas and suggestions are documented.
-------------------	-----	---

We're building the data infrastructure for large-scale carbon removal. This assignment is a small piece of what the job looks like. If you enjoy working through this, you'll enjoy the work.

Good luck.